CSci 23: Module 1 Applications Development →

Software & Applications

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Course Outcome

Manage application development using version control.

Learning Outcome

Identify the fundamental concepts about software.

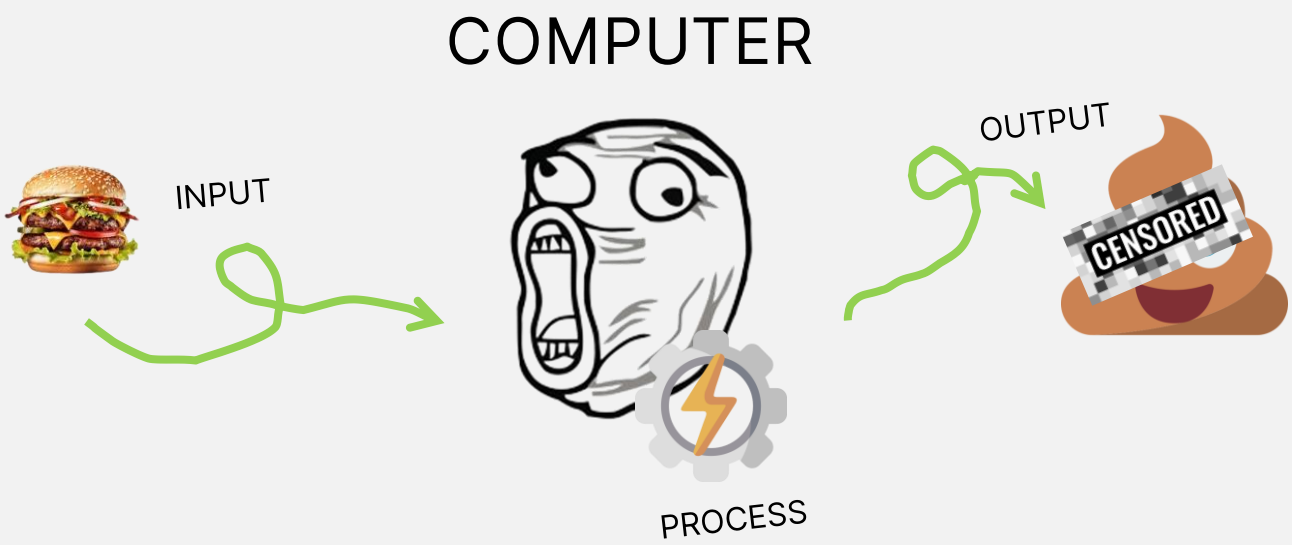
Outline

1. Software Concepts
2. Software Classifications
3. Software Development Lifecycle
4. Lifecycle Models

Software and Applications

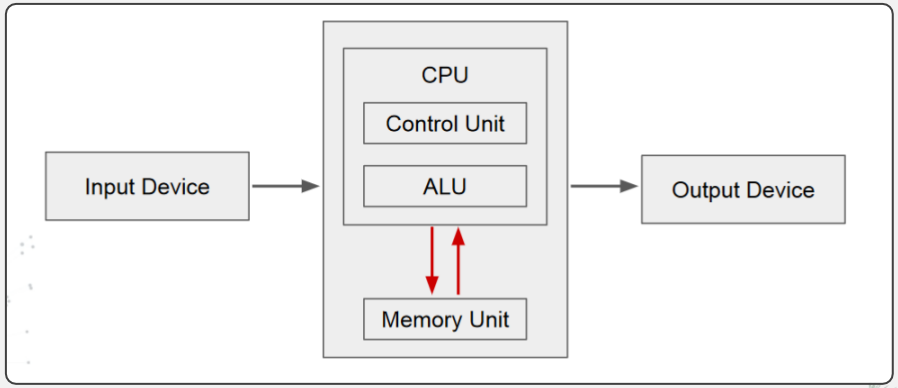
Software Concepts

What is a Computer?



What is a Computer?

A programmable electronic machine that accepts digitized input, processes according to a set of internally stored instructions, and produces the resulting output information.



What is a Computer?

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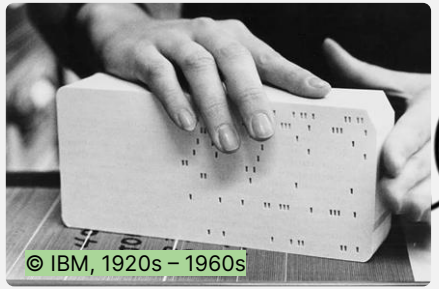
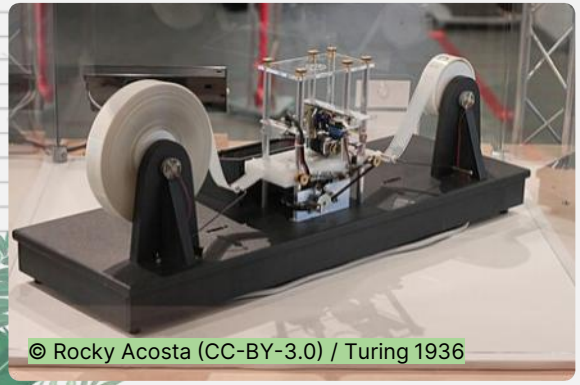


© IBM, 1920s – 1960s

"code"



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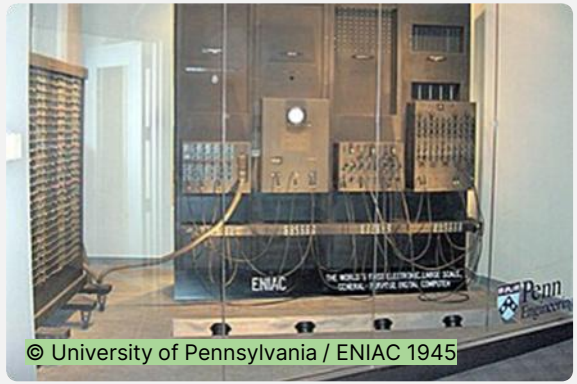

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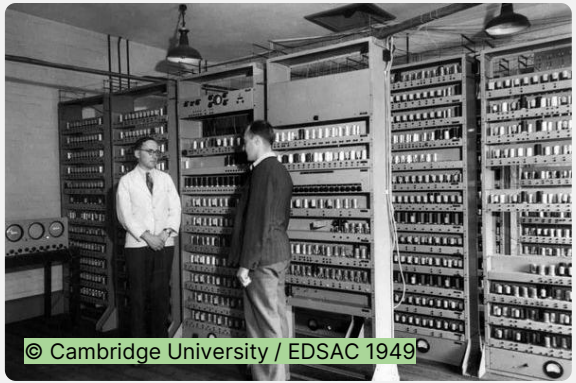

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Computer Classifications

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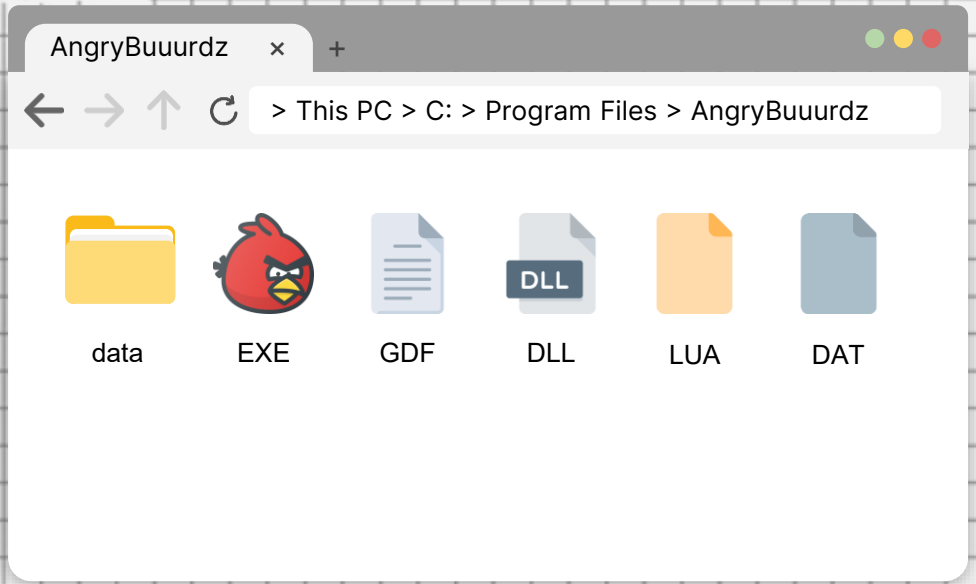
What is a Software?

It is a set of programs that enables computers to perform a specific task.

What is a Software?

- Open
- Run as administrator
- Open file location
- Properties



What is a Software?

Includes the following:

- set of programs,
- data and data structures,
- configuration files, and
- documentations detailing its design, development, operation, and maintenance.

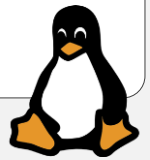
Software and Applications

Software Classifications

System Software

The manager of a general-purpose computer system and enables the execution of other applications.

- BIOS/UEFI, bootloader, firmware, operating systems, and device drivers.
- Memory management, concurrency, parallel computing, HW and low-level interfaces.
- Electrical, Electronics & Communication, and Computer engineers.



Firmware Bundle

A managed software bundle dedicated to a specialized HW and purpose, typically stored in a ROM or flash memory.

- BIOS/UEFI, bootloader, firmware, operating systems, and device drivers.
- Memory management, concurrency, parallel computing, HW and low-level interfaces.
- Electrical, Electronics & Communication, and Computer engineers.

Application Software (App)

Wide range of software used for specific tasks such as entertainment or productivity.

- Mobile, desktop, and web platforms.
- Knowledge in databases, high-level programming languages, and frameworks.
- Designers, IT, and Computer Science.

Business Software (Enterprise)

A sub-type of applications designed to assist businesses , including management, HR, accounting, inventory, and sales.

- Mobile, desktop, and web platforms.
- Knowledge in databases, high-level programming languages, and frameworks.
- Designers, IT/IS, and Computer Science.

Engineering and Scientific SW

Specialized for complex calculations, simulations, and data analysis with high accuracy and precision.

- Desktop and web platforms.
- Domain-specific expertise and software development skills.
- STEM, IT, and Computer Science.

Artificial Intelligence SW

Complex and hybrid systems to mimic or even outperform human intelligence and efficiency for various tasks.

- Mobile, desktop, and web platforms.
- Mathematical knowledge, domain-specific expertise, and software development skills.
- STEM, IT, and Computer Science.

Software and Applications

Software Development Lifecycle

What is Software Development?

The entire process from the initial ideation to the maintenance of a software.

* It requires skills on programming languages, libraries, frameworks, and relevant tools.

Planning

Implementation

Analysis

Testing

Maintenance

Design

Deployment

Software Development Lifecycle

It begins with the decision of starting a software project until delivery (IEEE-CS, 2014).

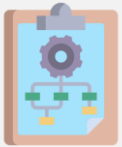
* The distinct phases have no specific order but often overlap or run in parallel.

Software and Applications

Lifecycle Models

Lifecycle Model

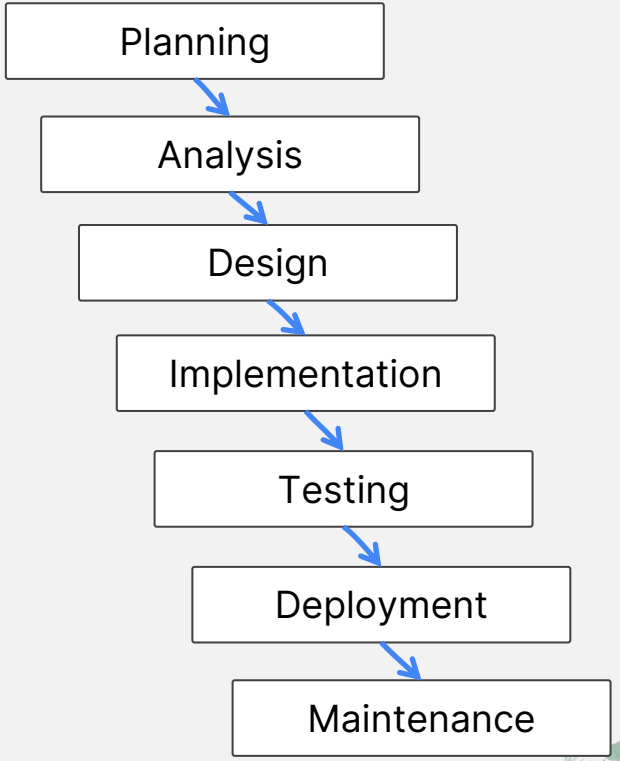
It provides the ordering of the phases of software development.



Waterfall Model

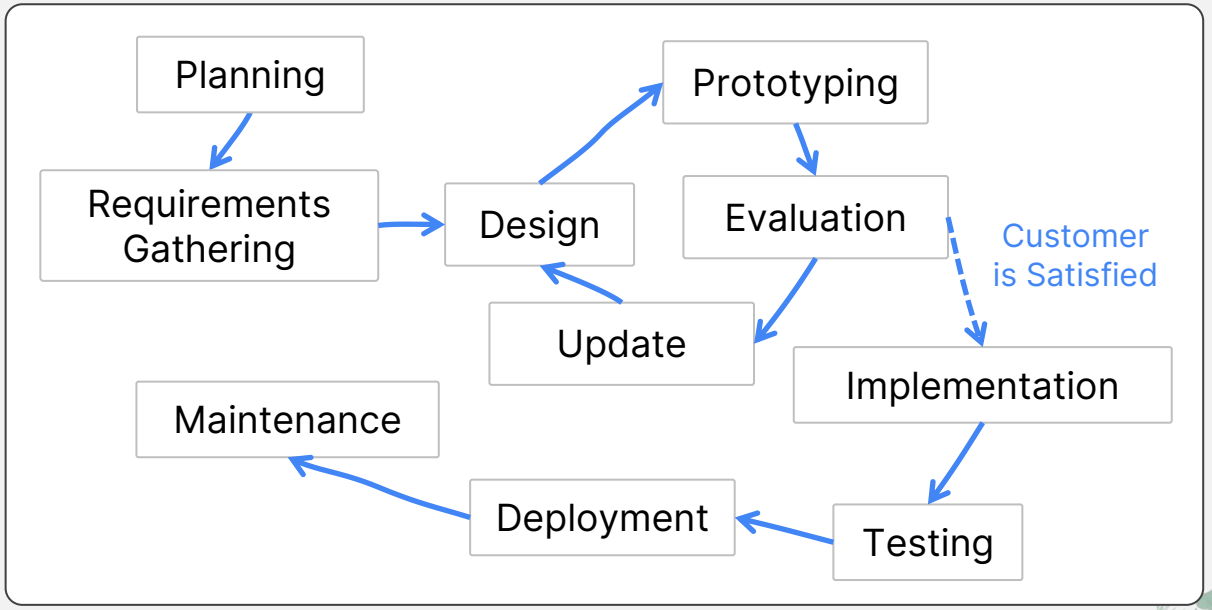
Software development done sequentially.

* Ideal to small software development done by those outside the field of computing.



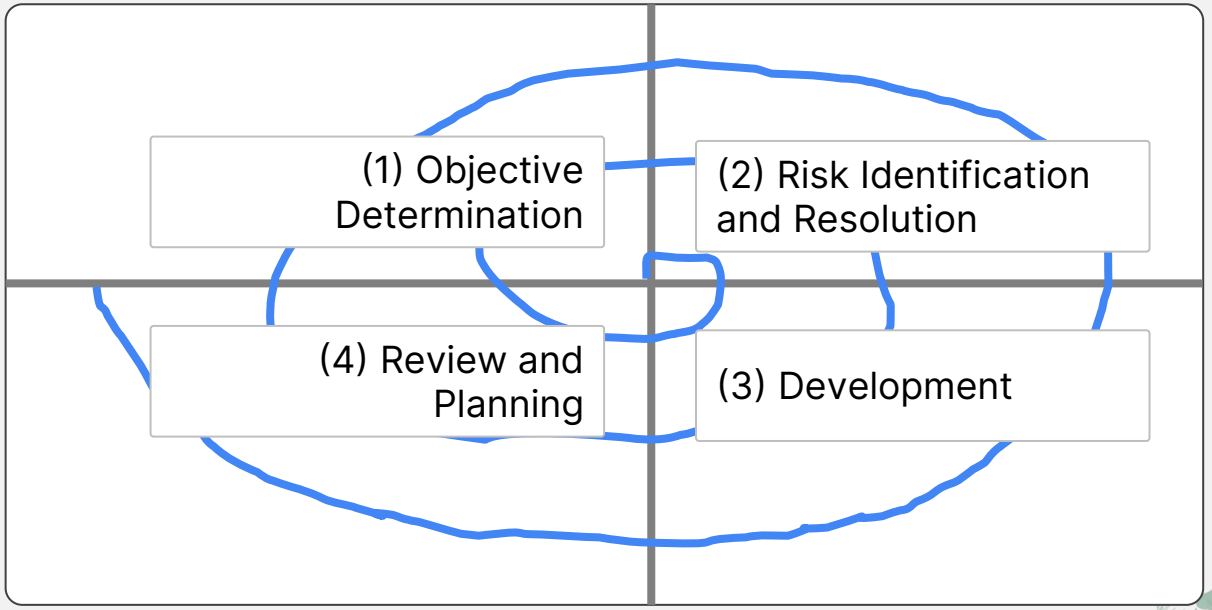
Prototype Model

It creates a preliminary version of the software before the final development.



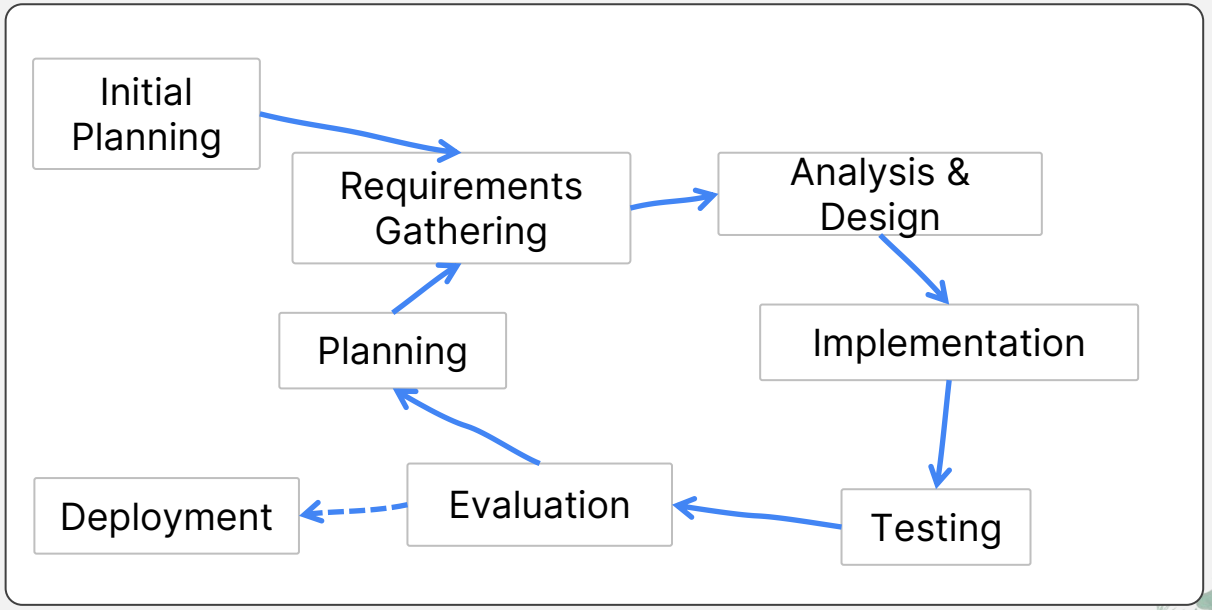
Spiral Model

It is a combination of sequential and prototype models, followed in a more defined manner.



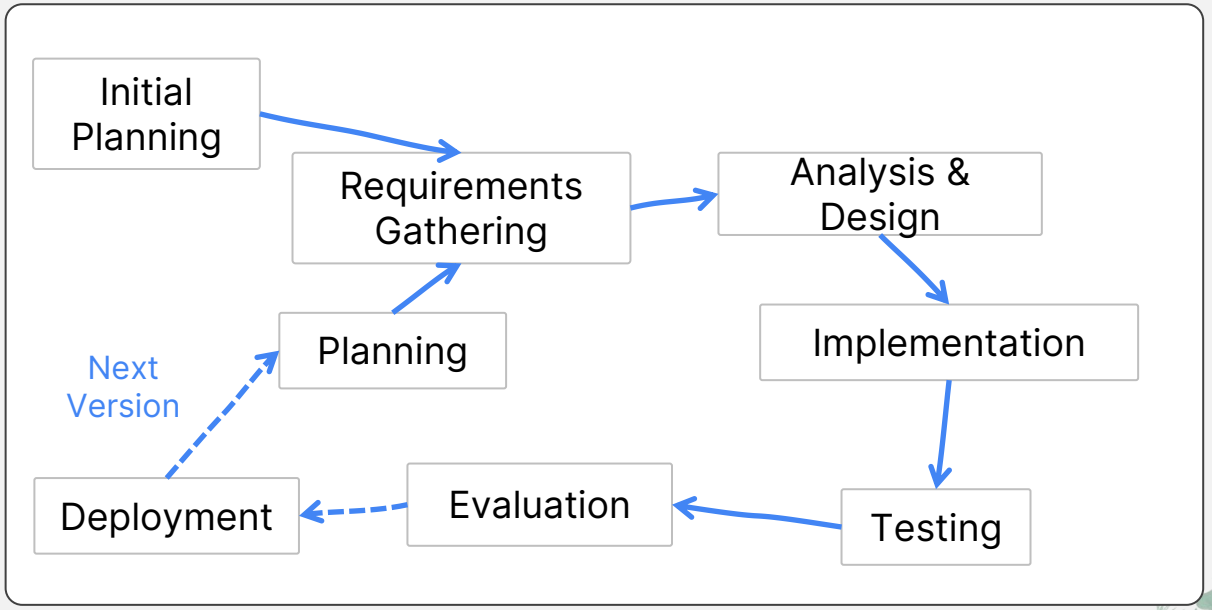
Iterative-Incremental Model

It starts with a basic implementation and adds features in each iteration until complete.



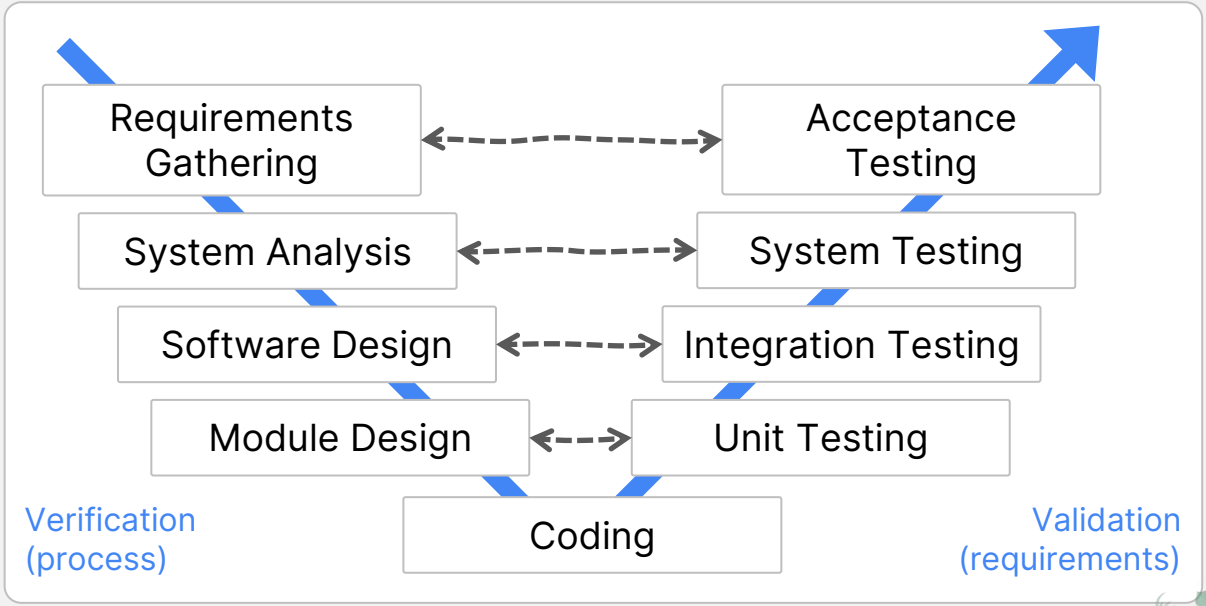
Evolutionary Model

It is like iterative-incremental model, but each iteration is a complete version.



V Model

It emphasizes on verification and validation through a structured approach.



Summary

- A software includes sets of programs, data and data structures, configuration files, and its documentations.
- We can classify software based on its usage and purpose.
- A software development lifecycle provides the different phases necessary in software development.
- A lifecycle model provides the arrangement of the distinct phases, including their order, overlap, and iteration.

Scan me!

Evaluation Form

forms.gle/TncdC5nEKJZYucVz6

Thank You!

References

- Bauer, F. L. (1972). Software Engineering. In Information Processing (p. 71).
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Eneeergy Check!

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Interested 🤔

2

Motivated!!!